Chapter 10: Canonical Correlation Analysis

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§ 10.1 Introduction

- How to describe the associations between two sets of variables.
- Canonical correlation analysis seeks to identify and quantify the associations between two sets of variables, which can effectively reveal the linear dependency relationship between two sets of variables.
- Canonical correlation analysis was initially proposed by H. Hotelling (1935,1936)

Another Example: Business Data Analysis

- Suppose you work for an e-commerce company and have two sets of data:
  - **Set X (User browsing behavior):** browsing time, number of clicks, page dwell time.
  - **Set Y (User purchasing outcomes):** purchase amount, number of items purchased, return/exchange frequency.

  You hope to know: What is the intrinsic relationship between a user's "browsing behavior pattern" and their "purchasing behavior pattern“.
- **Intuitive Results:**
  - **The first pair of canonical variables** might reveal that "rapid browsing + high clicks" is highly correlated with "low purchase amount + high returns" (correlation coefficient = 0.85). This uncovers an intuition: users who place rushed orders are more likely to regret their purchases.

Another Example: Business Data Analysis

- **Intuitive Results:**
  - **The second pair of canonical variables** might show that "long dwell time + low click frequency" is correlated with "high purchase amount + low item quantity." This reveals another insight: users who engage in deep comparison before ordering tend to buy high-value individual items.

- **Summary in One Sentence**

The core intuition of Canonical Correlation Analysis is: Within two sets of variables, construct a pair of "super variables" (linear combinations) such that the correlation between these two super variables is maximized, thereby revealing the strongest intrinsic structural relationship between the two sets. In essence, it seeks the most similar "projection directions" between two high-dimensional spaces.

§ 10.2 Population Canonical Correlations

- I) Definition and derivation of canonical correlations
- II) Property of canonical variables
- III) Calculating canonical correlations from correlation matrix

I) Definition and derivation of canonical correlation

- Suppose $\boldsymbol{x}=(x_1,x_2,\cdots,x_p)'$ and $\boldsymbol{y}=(y_1,y_2,\cdots,y_q)'$are two group of random variables, $Cov(\boldsymbol{x})=\boldsymbol{\Sigma}_{11}(>0)$, $Cov(\boldsymbol{y})=\boldsymbol{\Sigma}_{22}(>0)$, $\mathrm{Cov}(\boldsymbol{x}, \boldsymbol{y})=\boldsymbol{\Sigma}_{12}$. Then we have:

$$Cov\begin{pmatrix} \boldsymbol{x} \\ \boldsymbol{y} \end{pmatrix} = \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}$$

where $\boldsymbol{\Sigma}_{21}=\boldsymbol{\Sigma}_{12}'$.

- How to describe the correlations between $\boldsymbol{x}$ and $\boldsymbol{y}$?

We transfer this question to study the correlation between $u=\boldsymbol{a}'\boldsymbol{x}$ and $v=\boldsymbol{b}'\boldsymbol{y}$, where $\boldsymbol{a}=(a_1,a_2,\cdots,a_p)'$, $\boldsymbol{b}=(b_1,b_2,\cdots,b_q)'$.

So, we need to calculate

$$\mathrm{Cov}(u,v)=\mathrm{Cov}(\boldsymbol{a}'\boldsymbol{x},\boldsymbol{b}'\boldsymbol{y})=\boldsymbol{a}'\mathrm{Cov}(\boldsymbol{x},\boldsymbol{y})\boldsymbol{b}=\boldsymbol{a}'\boldsymbol{\Sigma}_{12}\boldsymbol{b}$$

$$V(u)=V(\boldsymbol{a}'\boldsymbol{x})=\boldsymbol{a}'V(\boldsymbol{x})\boldsymbol{a}=\boldsymbol{a}'\boldsymbol{\Sigma}_{11}\boldsymbol{a}$$

$$V(v)=V(\boldsymbol{b}'\boldsymbol{y})=\boldsymbol{b}'V(\boldsymbol{y})\boldsymbol{b}=\boldsymbol{b}'\boldsymbol{\Sigma}_{22}\boldsymbol{b}$$

I) Definition and derivation of canonical correlation

- Therefore, the correlation between u and v is defined as

$$\rho(u,v)=\frac{\boldsymbol{a}'\boldsymbol{\Sigma}_{12}\boldsymbol{b}}{\sqrt{\boldsymbol{a}'\boldsymbol{\Sigma}_{11}\boldsymbol{a}}\sqrt{\boldsymbol{b}'\boldsymbol{\Sigma}_{22}\boldsymbol{b}}} \tag{10.2.3}$$

- Under the constrains:

$$V(u)=1,\quad V(v)=1 \tag{10.2.4}$$

that is

$$\boldsymbol{a}'\boldsymbol{\Sigma}_{11}\boldsymbol{a}=1,\quad \boldsymbol{b}'\boldsymbol{\Sigma}_{22}\boldsymbol{b}=1, \tag{10.2.5}$$

we hope to find some $\boldsymbol{a}\in R^p$ some $\boldsymbol{b}\in R^q$ such that

$$\rho(u,v)=\boldsymbol{a}'\boldsymbol{\Sigma}_{12}\boldsymbol{b} \tag{10.2.6}$$

achieves the maximum.

I) Definition and derivation of canonical correlation

i) $\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}, \boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}, \boldsymbol{\Sigma}_{11}^{-1/2}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1/2}(\geq 0)$ and $\boldsymbol{\Sigma}_{22}^{-1/2}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1/2}(\geq 0)$ have the same non-zero eigenvalues $\rho_1^2 \geq \rho_2^2 \geq \cdots \geq \rho_m^2 > 0$ with m being the rank of $\boldsymbol{\Sigma}_{12}$.

ii) From $rank(AA') = rank(A'A) = rank(A)$, we have

$$\begin{aligned}\operatorname{rank}\left(\boldsymbol{\Sigma}_{22}^{-1/2}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1/2}\right) &= \operatorname{rank}\left(\boldsymbol{\Sigma}_{11}^{-1/2}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1/2}\right)\\ &= \operatorname{rank}\left(\boldsymbol{\Sigma}_{12}\right) = m\end{aligned}$$

- ρ_i is the square root of ρ_i^2, $i = 1,2,\cdots,m$.

Let $\beta_1, \beta_2, \cdots, \beta_m$ be the m positive orthonormal eigenvectors of $\Sigma_{22}^{-1/2}\Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}\Sigma_{22}^{-1/2}$ corresponding to eigenvalues $\rho_1^2, \rho_2^2, \cdots, \rho_m^2$.

Denote by $\alpha_i = \Sigma_{11}^{-1/2}\Sigma_{12}\Sigma_{22}^{-1/2}\beta_i/\rho_i$, $a_i = \Sigma_{11}^{-1/2}\alpha_i$, $b_i = \Sigma_{22}^{-1/2}\beta_i$.

Then, the following conclusions holds.

Proposition 1 (Exer 1):

- $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \cdots, \boldsymbol{\alpha}_m$ are orthonormal eigenvectors of $\Sigma_{11}^{-1/2}\Sigma_{12}\Sigma_{22}^{-1/2}\Sigma_{21}$ corresponding to $\rho_1^2, \rho_2^2, \cdots, \rho_m^2$.
- $\boldsymbol{a}_1, \boldsymbol{a}_2, \cdots, \boldsymbol{a}_m$ are eigenvectors of $\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$ corresponding to $\rho_1^2, \rho_2^2, \cdots, \rho_m^2$.
- $\boldsymbol{b}_1, \boldsymbol{b}_2, \cdots, \boldsymbol{b}_m$ are eigenvectors of $\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}$ corresponding to $\rho_1^2, \rho_2^2, \cdots, \rho_m^2$.

Problem: For $a \in R^p, b \in R^q$, to find

$$max_{a,b}\rho(u,v) = max_{a,b}a'\Sigma_{12}b \qquad (10.2.6)$$

under the constraints of $a'\Sigma_{11}a = 1,\ b'\Sigma_{22}b.$

Proof: Let $\alpha = \Sigma_{11}^{1/2}a$, $\beta = \Sigma_{22}^{1/2}b$. Then, the above constraints are equivalent to $\alpha'\alpha = 1$ and $\beta'\beta = 1$. Using Cauchy inequality, we obtain

$$\begin{aligned}(a'\Sigma_{12}b)^2 &= (\alpha'\Sigma_{11}^{-1/2}\Sigma_{12}\Sigma_{22}^{-1/2}\beta)^2 \\ &\le (\alpha'\alpha)\left[\left(\Sigma_{11}^{-\frac{1}{2}}\Sigma_{12}\Sigma_{22}^{-\frac{1}{2}}\beta\right)'\left(\Sigma_{11}^{-\frac{1}{2}}\Sigma_{12}\Sigma_{22}^{-\frac{1}{2}}\beta\right)\right] \\ &= \beta'\,\Sigma_{22}^{-1/2}\Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}\Sigma_{22}^{-1/2}\beta. \qquad (10.2.7)\end{aligned}$$

From (1.8.3), it follows that (10.2.7) holds when $\beta = \beta_1$.

Correspondingly, $\alpha = \alpha_1 = \frac{1}{\rho_1}\Sigma_{11}^{-1/2}\Sigma_{12}\Sigma_{22}^{-1/2}\beta_1$ (unit vector) .

Therefore, when $a = a_1 = \Sigma_{11}^{-1/2}\alpha_1, b = b_1 = \Sigma_{22}^{-1/2}\beta_1, \rho(u,v) = a'\Sigma_{12}b$ attains the maximum $\rho_1 = \alpha_1'\,\Sigma_{11}^{-1/2}\Sigma_{12}\Sigma_{22}^{-1/2}\beta_1$.

When taking $\boldsymbol{a} = \boldsymbol{a}_1$, $\boldsymbol{b} = \boldsymbol{b}_1$, condition (10.2.5) is satisfied, and $\rho(u,v)=\boldsymbol{a}'\boldsymbol{\Sigma}_{12}\boldsymbol{b}$ achieves maximum ρ_1. Obviously $\rho_1 \le 1$. Denote

$$u_1 = \boldsymbol{a}_1'\boldsymbol{x}, \quad v_1 = \boldsymbol{b}_1'\boldsymbol{y}$$

as the first pair of canonical variables (第一对典型相关变量),

denote $\boldsymbol{a}_1$, $\boldsymbol{b}_1$ as the first pair of canonical correlation vectors,

denote ρ_1 as the first canonical correlation coefficient (第一个典型相关系数).

- The first pair of canonical variables u_1, v_1 captures the primary correlation between between $\boldsymbol{x}$ and $\boldsymbol{y}$. If additional correlation remains, we can search for a second pair of canonical variables $u_2=\boldsymbol{a}'\boldsymbol{x}$, $v_2=\boldsymbol{b}'\boldsymbol{y}$, that is, $\boldsymbol{a}$, $\boldsymbol{b}$ must meet standard conditions (10.2.5), ensuring that the second pair does not duplicate information contained in the first pair.

$$\rho(u_2,u_1)=\rho(\boldsymbol{a}'\boldsymbol{x},\boldsymbol{a}_1'\boldsymbol{x})=\mathrm{Cov}(\boldsymbol{a}'\boldsymbol{x},\boldsymbol{a}_1'\boldsymbol{x})=\boldsymbol{a}'\boldsymbol{\Sigma}_{11}\boldsymbol{a}_1=0;$$

$$\rho(v_2,v_1)=\rho(\boldsymbol{b}'\boldsymbol{y},\boldsymbol{b}_1'\boldsymbol{y})=\mathrm{Cov}(\boldsymbol{b}'\boldsymbol{y},\boldsymbol{b}_1'\boldsymbol{y})=\boldsymbol{b}'\boldsymbol{\Sigma}_{22}\boldsymbol{b}_1=0.$$

Subject to the constraints (10.2.5), $\boldsymbol{a}'\boldsymbol{\Sigma}_{11}\boldsymbol{a}_1=0$ and $\boldsymbol{b}'\boldsymbol{\Sigma}_{22}\boldsymbol{b}_1=0$, we aim to maximize

$$\rho(u_2,v_2)=\rho(\boldsymbol{a}'\boldsymbol{x},\boldsymbol{b}'\boldsymbol{y})=\boldsymbol{a}'\boldsymbol{\Sigma}_{12}\boldsymbol{b}.$$

- Generally, the i-th$(1 \le i \le m)$ pair of canonical variables(第 i对典型相关变量) $u_i=\boldsymbol{a}'\boldsymbol{x}$, $v_i=\boldsymbol{b}'\boldsymbol{y}$ is determined by finding $\boldsymbol{a}\in R^p$, $\boldsymbol{b}\in R^q$ under the constrains

$$\boldsymbol{a}'\boldsymbol{\Sigma}_{11}\boldsymbol{a}=1,\ \boldsymbol{b}'\boldsymbol{\Sigma}_{22}\boldsymbol{b}=1 \tag{10.2.5}$$

$$\boldsymbol{a}'\boldsymbol{\Sigma}_{11}\boldsymbol{a}_k=0,\ \boldsymbol{b}'\boldsymbol{\Sigma}_{22}\boldsymbol{b}_k=0,\quad k=1,2,\cdots,i-1, \tag{10.2.10}$$

such that

$$\rho(u_i,v_i)=\rho(\boldsymbol{a}'\boldsymbol{x},\boldsymbol{b}'\boldsymbol{y})=\boldsymbol{a}'\boldsymbol{\Sigma}_{12}\boldsymbol{b} \tag{10.2.11}$$

achieves the maximum ρ_i.

- When $\boldsymbol{a}=\boldsymbol{a}_i$, $\boldsymbol{b}=\boldsymbol{b}_i$, $\rho(u_i,v_i)$ achieves the maximum ρ_i under constrains (10.2.5) and (10.2.10), where ρ_i is referred to the i-th canonical correlation coefficient(第i个典型相关系数), and $\boldsymbol{a}_i$, $\boldsymbol{b}_i$ are called as the i-th pair of canonical correlation vectors or coefficients (第i对典型相关向量或第i对典型系数).

II) Property of canonical variables

- 1. Uncorrelation of the canonical variables within the group
- 2. Correlation of canonical variables among groups
- 3. Correlation between original variables and canonical variables
- 4. Canonical correlation is some kind of multiple correlation coefficient
- 5.Connection between correlation, multiple correlation and canonical correlation

1. Uncorrelation of the canonical variables within the group

- Suppose the i-th pair of canonical variables of $\boldsymbol{x}$, $\boldsymbol{y}$ is

$$u_i=\boldsymbol{a}_i'\boldsymbol{x},\ v_i=\boldsymbol{b}_i'\boldsymbol{y},\quad i=1,2,\cdots,m. \tag{10.2.13}$$

Then we have

$$V(u_i)=\boldsymbol{a}_i'\boldsymbol{\Sigma}_{11}\boldsymbol{a}_i=1, V(v_i)=\boldsymbol{b}_i'\boldsymbol{\Sigma}_{22}\boldsymbol{b}_i=1,\ i=1,2,\cdots,m$$

$$\begin{aligned}\rho(u_i,u_j)&=\mathrm{Cov}(u_i,u_j)=\boldsymbol{a}_i'\boldsymbol{\Sigma}_{11}\boldsymbol{a}_j=0,\ 1\le i\ne j\le m\\ \rho(v_i,v_j)&=\mathrm{Cov}(v_i,v_j)=\boldsymbol{b}_i'\boldsymbol{\Sigma}_{22}\boldsymbol{b}_j=0,\ 1\le i\ne j\le m\end{aligned} \tag{10.2.14}$$

2. Correlation of canonical variables among groups

- the i-th canonical correlation coefficient

$$\rho(u_i,v_i)=\rho_i\,,i=1,2,\cdots,m, \qquad (10.2.15)$$

$$\rho(u_i,v_j)=\mathrm{Cov}(u_i,v_j)=\mathrm{Cov}(\boldsymbol{a}_i'\boldsymbol{x},\boldsymbol{b}_j'\boldsymbol{y})=\boldsymbol{a}_i'\mathrm{Cov}(\boldsymbol{x},\boldsymbol{y})\boldsymbol{b}_j$$

$$=\boldsymbol{\alpha}_i'\boldsymbol{\Sigma}_{11}^{-1/2}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1/2}\boldsymbol{\beta}_j=\rho_j\boldsymbol{\alpha}_i'\boldsymbol{\alpha}_j=0, 1\le i\ne j\le m \qquad (10.2.16)$$

since $\beta_j=\rho_j\Sigma_{22}^{1/2}\Sigma_{12}^{-1}\Sigma_{11}^{1/2}\alpha_j$ from the previous definition.

- Let $\boldsymbol{u}=(u_1,u_2,\cdots,u_m)'$ and $\boldsymbol{v}=(v_1,v_2,\cdots,v_m)'$, then these two properties can be represented by matrices as

$$Cov(\boldsymbol{u})=\boldsymbol{I},\ Cov(\boldsymbol{v})=\boldsymbol{I},\ \mathrm{Cov}(\boldsymbol{u},\boldsymbol{v})=\boldsymbol{\Lambda} \text{ or}$$

$$Cov\begin{pmatrix}\boldsymbol{u}\\ \boldsymbol{v}\end{pmatrix}=\begin{pmatrix}\boldsymbol{I} & \boldsymbol{\Lambda}\\ \boldsymbol{\Lambda} & \boldsymbol{I}\end{pmatrix}$$

where $\boldsymbol{\Lambda}=\mathrm{diag}(\rho_1,\rho_2,\cdots,\rho_m)$.

3. Correlation between original variables and canonical variables

- Let $\boldsymbol{A}=(\boldsymbol{a}_1,\boldsymbol{a}_2,\cdots,\boldsymbol{a}_m)$, $\boldsymbol{B}=(\boldsymbol{b}_1,\boldsymbol{b}_2,\cdots,\boldsymbol{b}_m)$, then

$$\boldsymbol{u}=\boldsymbol{A}'\boldsymbol{x}, \quad \boldsymbol{v}=\boldsymbol{B}'\boldsymbol{y}$$

- The covariance matrices between original variables and canonical variables are formulated as

$$\begin{aligned}
\mathrm{Cov}(\boldsymbol{x},\boldsymbol{u})&=\mathrm{Cov}(\boldsymbol{x},\boldsymbol{A}'\boldsymbol{x})=\boldsymbol{\Sigma}_{11}\boldsymbol{A}\\
\mathrm{Cov}(\boldsymbol{x},\boldsymbol{v})&=\mathrm{Cov}(\boldsymbol{x},\boldsymbol{B}'\boldsymbol{y})=\boldsymbol{\Sigma}_{12}\boldsymbol{B}\\
\mathrm{Cov}(\boldsymbol{y},\boldsymbol{u})&=\mathrm{Cov}(\boldsymbol{y},\boldsymbol{A}'\boldsymbol{x})=\boldsymbol{\Sigma}_{21}\boldsymbol{A}\\
\mathrm{Cov}(\boldsymbol{y},\boldsymbol{v})&=\mathrm{Cov}(\boldsymbol{y},\boldsymbol{B}'\boldsymbol{y})=\boldsymbol{\Sigma}_{22}\boldsymbol{B}
\end{aligned} \qquad (10.2.17)$$

Properties of Covariance Matrix:

(2) Let $\boldsymbol{A}$ be a constant matrix, $\boldsymbol{b}$ is a constant vector, then

$$Cov(\mathbf{Ax}+\mathbf{b})=\mathbf{A}Cov(\mathbf{x})\mathbf{A}'$$

4. Correlation between original variables and canonical variables

$$
\begin{aligned}
\rho(\boldsymbol{x},\boldsymbol{u}) &= \boldsymbol{D}_1^{-1}\boldsymbol{\Sigma}_{11}\boldsymbol{A}, \\
\rho(\boldsymbol{x},\boldsymbol{v}) &= \boldsymbol{D}_1^{-1}\boldsymbol{\Sigma}_{12}\boldsymbol{B} \\
\rho(\boldsymbol{y},\boldsymbol{u}) &= \boldsymbol{D}_2^{-1}\boldsymbol{\Sigma}_{21}\boldsymbol{A}, \\
\rho(\boldsymbol{y},\boldsymbol{v}) &= \boldsymbol{D}_2^{-1}\boldsymbol{\Sigma}_{22}\boldsymbol{B}
\end{aligned}
\qquad (10.2.18)
$$

where

$$\boldsymbol{D}_1 = \mathrm{diag}\left(\sqrt{V(x_1)},\cdots,\sqrt{V(x_p)}\right),$$

$$\boldsymbol{D}_2 = \mathrm{diag}\left(\sqrt{V(y_1)},\cdots,\sqrt{V(y_q)}\right)$$

Proof of (10.2.18): We will prove the first equation. The other three equations can be proved similarly. Let

$$\boldsymbol{x}^* = \boldsymbol{D}_1^{-1}\left(\boldsymbol{x} - \boldsymbol{\mu}_1\right), \quad \boldsymbol{y}^* = \boldsymbol{D}_2^{-1}\left(\boldsymbol{y} - \boldsymbol{\mu}_2\right), \quad (10.2.19)$$

where $\boldsymbol{\mu}_1 = E(\boldsymbol{x})$, $\boldsymbol{\mu}_2 = E(\boldsymbol{y})$, that is, we do standard transformation of the component of x and y, then we have

$$\rho\left(\boldsymbol{x}, \boldsymbol{u}\right) = \rho\left(\boldsymbol{x}^*, \boldsymbol{u}\right) = \mathrm{Cov}\left(\boldsymbol{x}^*, \boldsymbol{u}\right)$$

$$= \mathrm{Cov}\left[\boldsymbol{D}_1^{-1}\left(\boldsymbol{x} - \boldsymbol{\mu}_1\right), \boldsymbol{u}\right] = \boldsymbol{D}_1^{-1}\mathrm{Cov}\left(\boldsymbol{x}, \boldsymbol{u}\right).$$

$$= \boldsymbol{D}_1^{-1}\boldsymbol{\Sigma}_{11}\boldsymbol{A}$$

4. Canonical correlation is some kind of multiple correlation coefficient

Exercise 2

- The multiple correlation coefficient between $u_i = \boldsymbol{a}_i'\boldsymbol{x}$ and $\boldsymbol{y}$ is

$$\rho_{u_i \bullet \boldsymbol{y}} = \rho_i, \quad i = 1, 2, \cdots, m \qquad (10.2.20)$$

Hints: $\rho_{u_i \cdot y}^2 = \frac{Cov(u_i, y)\Sigma_{22}^{-1}Cov(y, u_i)}{V(u_i)} = \rho_i^2$.

- The multiple correlation coefficient between $v_i = \boldsymbol{b}_i'\boldsymbol{y}$ and $\boldsymbol{y}$ is

$$\rho_{v_i \bullet \boldsymbol{x}} = \rho_i, \quad i = 1, 2, \cdots, m \qquad (10.2.21)$$

Hints: $\rho_{v_i \cdot x}^2 = \frac{Cov(v_i, x)\Sigma_{11}^{-1}Cov(x, v_i)}{V(v_i)} = \rho_i^2$.

5.Connection between correlation, multiple correlation and canonical correlation

- When $p=q=1$, the (only) canonical correlation of $\boldsymbol{x}$ and $\boldsymbol{y}$ is their correlation.
- When $p=1$or $q=1$, the (only) canonical correlation of $\boldsymbol{x}$ and $\boldsymbol{y}$ is their multiple correlation.
- Therefore, multiple correlation is an example of canonical correlation, and correlation is an example of multiple correlation.
- The first canonical correlation is no less than the multiple correlation coefficient between any component of $\boldsymbol{x}$ (or y) and $\boldsymbol{y}$ (or $\boldsymbol{x}$), even if these multiple correlation coefficients are small, the first canonical correlation could still be large.
- Similarly, when $p=1$or $q=1$, the multiple correlation of $\boldsymbol{x}$ (or $\boldsymbol{y}$) and $\boldsymbol{y}$ (or $\boldsymbol{x}$) will be no less than the correlation between any component of $\boldsymbol{y}$ and $\boldsymbol{x}$, even if these correlations is small, the multiple correlation could still be large.

III) Calculating canonical correlations from correlation matrix

- Sometimes, the unit of the components of $\boldsymbol{x}$ and $\boldsymbol{y}$ differ, so we aim to standardize them before conducting canonical correlation analysis.
- Suppose $\boldsymbol{R}=\begin{pmatrix}\boldsymbol{R}_{11} & \boldsymbol{R}_{12}\\ \boldsymbol{R}_{21} & \boldsymbol{R}_{22}\end{pmatrix}$ is the correlation matrix of $\begin{pmatrix}\boldsymbol{x}\\ \boldsymbol{y}\end{pmatrix}$, now

we want to calculate the canonical variables of $\boldsymbol{x}^*$ and $\boldsymbol{y}^*$:

$$\boldsymbol{a}_i^{*\prime}\boldsymbol{x}^*,\ \boldsymbol{b}_i^{*\prime}\boldsymbol{y}^*,\quad i=1,2,\cdots,m$$

$$V\left(\boldsymbol{x}^*\right)=\boldsymbol{D}_1^{-1}V\left(\boldsymbol{x}\right)\boldsymbol{D}_1^{-1}=\boldsymbol{D}_1^{-1}\boldsymbol{\Sigma}_{11}\boldsymbol{D}_1^{-1}=\boldsymbol{R}_{11}$$

$$V\left(\boldsymbol{y}^*\right)=\boldsymbol{D}_2^{-1}V\left(\boldsymbol{y}\right)\boldsymbol{D}_2^{-1}=\boldsymbol{D}_2^{-1}\boldsymbol{\Sigma}_{22}\boldsymbol{D}_2^{-1}=\boldsymbol{R}_{22}$$

$$\mathrm{Cov}\left(\boldsymbol{x}^*,\boldsymbol{y}^*\right)=\boldsymbol{D}_1^{-1}\mathrm{Cov}\left(\boldsymbol{x},\boldsymbol{y}\right)\boldsymbol{D}_2^{-1}=\boldsymbol{D}_1^{-1}\boldsymbol{\Sigma}_{12}\boldsymbol{D}_2^{-1}=\boldsymbol{R}_{12}$$

$$\mathrm{Cov}\left(y^*,x^*\right)=\boldsymbol{D}_2^{-1}\mathrm{Cov}\left(\boldsymbol{y},\boldsymbol{x}\right)\boldsymbol{D}_1^{-1}=\boldsymbol{D}_2^{-1}\boldsymbol{\Sigma}_{21}\boldsymbol{D}_1^{-1}=\boldsymbol{R}_{21}$$

then we have

$$R_{11}^{-1}R_{12}R_{22}^{-1}R_{21} = \left(D_1^{-1}\Sigma_{11}D_1^{-1}\right)^{-1} D_1^{-1}\Sigma_{12}D_2^{-1}\left(D_2^{-1}\Sigma_{22}D_2^{-1}\right)^{-1} D_2^{-1}\Sigma_{21}D_1^{-1}.$$

$$= D_1\Sigma_{11}^{-1}\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}D_1^{-1}$$

since

$$\Sigma_{11}^{-1}\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}a_i = \rho_i^2 a_i$$

$$D_1\Sigma_{11}^{-1}\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}D_1^{-1}\left(D_1 a_i\right) = \rho_i^2\left(D_1 a_i\right)$$

therefore,

$$R_{11}^{-1}R_{12}R_{22}^{-1}R_{21}a_i^* = \rho_i^2 a_i^*. \qquad (10.2.22)$$

where $a_i^* = D_1 a_i$ and $a_i^{*\prime}R_{11}a_i^* = a_i' D_1 R_{11} D_1 a_i = a_i'\Sigma_{11}a_i = 1$.

Similarly

$$R_{22}^{-1}R_{21}R_{11}^{-1}R_{12}b_i^* = \rho_i^2 b_i^*. \qquad (10.2.22')$$

where $b_i^* = D_2 b_i$ and $b_i^{*\prime}R_{22}b_i^* = b_i' D_2 R_{22} D_2 b_i = b_i'\Sigma_{22}b_i = 1$.

Let $R = \begin{pmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{pmatrix}$ be the correlation matrix of $\begin{pmatrix} x \\ y \end{pmatrix}$.

$D_1 = diag(\sqrt{v(x_1)}, \cdots, \sqrt{v(x_p)}), D_2 = diag(\sqrt{v(y_1)}, \cdots, \sqrt{v(y_q)})$

Denote by $a_i^* = D_1 a_i$ such that $a_i^{*\prime} R_{11} a_i' = 1$, and $b_i^* = D_2 b_i$ such that $b_i^{*\prime} R_{22} b_i' = 1$. Then, the following conclusions holds.

Proposition 2

- $a_1^*, a_2^*, \cdots, a_m^*$ are eigenvectors of $R_{11}^{-1} R_{12} R_{22}^{-1} R_{21}$ corresponding to $\rho_1^2, \rho_2^2, \cdots, \rho_m^2$.
- $b_1^*, b_2^*, \cdots, b_m^*$ are eigenvectors of $R_{22}^{-1} R_{21} R_{11}^{-1} R_{12}$ corresponding to $\rho_1^2, \rho_2^2, \cdots, \rho_m^2$.

Proposition 2

- Therefore, from the definition of canonical coefficients, we know that $\boldsymbol{a}_i^*, \boldsymbol{b}_i^*$ are the i -th canonical correlation vectors of $\boldsymbol{x}^*$ and $\boldsymbol{y}^*$, $u_i^* = a_i^{*\prime} x^*$, $v_i^* = b_i^{*\prime} y^*$ is the i-th$(1 \le i \le m)$ pair of canonical variables(第 i对典型相关变量) and its i-th canonical correlation $\rho(u_i^*, v_i^*)$ is still ρ_i. The canonical correlations ρ_i are unchanged under standardization.
- Since

$$u_i^* = \boldsymbol{a}_i^{*\prime} \boldsymbol{x}^* = \boldsymbol{a}_i' \boldsymbol{D}_1 \boldsymbol{D}_1^{-1}\left(\boldsymbol{x} - \boldsymbol{\mu}_1\right) = \boldsymbol{a}_i' \boldsymbol{x} - \boldsymbol{a}_i' \boldsymbol{\mu}_1 = u_i - \boldsymbol{a}_i' \boldsymbol{\mu}_1$$

$$v_i^* = \boldsymbol{b}_i^{*\prime} \boldsymbol{y}^* = \boldsymbol{b}_i' \boldsymbol{D}_2 \boldsymbol{D}_2^{-1}\left(\boldsymbol{y} - \boldsymbol{\mu}_2\right) = \boldsymbol{b}_i' \boldsymbol{y} - \boldsymbol{b}_i' \boldsymbol{\mu}_2 = v_i - \boldsymbol{b}_i' \boldsymbol{\mu}_2$$

So the i-th pair of canonical variables $u_i^* = \boldsymbol{a}_i^{*\prime} \boldsymbol{x}^*, v_i^* = \boldsymbol{b}_i^{*\prime} \boldsymbol{y}^*$ are the centered value of $u_i = \boldsymbol{a}_i' \boldsymbol{x}$, $v_i = \boldsymbol{b}_i' \boldsymbol{y}$, and hence have mean zero.

Example 10.2.1 Suppose ***x***, ***y*** has the following correlation matrix:

$$\boldsymbol{R}_{11} = \begin{pmatrix} 1 & \alpha \\ \alpha & 1 \end{pmatrix}, \quad \boldsymbol{R}_{22} = \begin{pmatrix} 1 & \gamma \\ \gamma & 1 \end{pmatrix}, \quad \boldsymbol{R}_{12} = \begin{pmatrix} \beta & \beta \\ \beta & \beta \end{pmatrix} = \beta \mathbf{1}\mathbf{1}'$$

where $|\alpha| < 1, |\gamma| < 1$, we can guarantee that $\boldsymbol{R}_{11}^{-1}, \boldsymbol{R}_{22}^{-1}$ exist.

Try to find the first pair of canonical variables and the first canonical correlation for ***x***, ***y***.

Solution: $$\boldsymbol{R}_{11}^{-1}\boldsymbol{R}_{12}\boldsymbol{R}_{22}^{-1}\boldsymbol{R}_{21}=\frac{1}{1-\alpha^2}\begin{pmatrix}1 & -\alpha\\ -\alpha & 1\end{pmatrix}\beta\mathbf{1}\mathbf{1}'\bullet\frac{1}{1-\gamma^2}\begin{pmatrix}1 & -\gamma\\ -\gamma & 1\end{pmatrix}\beta\mathbf{1}\mathbf{1}'$$

$$=\frac{\beta^2}{\left(1-\alpha^2\right)\left(1-\gamma^2\right)}\begin{pmatrix}1-\alpha\\ 1-\alpha\end{pmatrix}\left(1-\gamma,1-\gamma\right)\mathbf{1}\mathbf{1}'$$

$$=\frac{\beta^2}{(1+\alpha)(1+\gamma)}\mathbf{1}\mathbf{1}'\mathbf{1}\mathbf{1}'=\frac{2\beta^2}{(1+\alpha)(1+\gamma)}\mathbf{1}\mathbf{1}'$$

Since $\mathbf{1}\mathbf{1}'$ has only non-zero eigenvalue $\mathbf{1}'\mathbf{1}$=2, so $\boldsymbol{R}_{11}^{-1}\boldsymbol{R}_{12}\boldsymbol{R}_{22}^{-1}\boldsymbol{R}_{21}$ has the only non-zero eigenvalue

$$\rho_1^2=\frac{4\beta^2}{(1+\alpha)(1+\gamma)}.$$

- Under constrain $\boldsymbol{a}_1^{*\prime}\boldsymbol{R}_{11}\boldsymbol{a}_1^*=1$, the eigenvector corresponding to eigenvalue ρ_1^2 is $\boldsymbol{a}_1^*=\left[2(1+\alpha)\right]^{-1/2}\mathbf{1}$.

Similarly, under the constrain $\boldsymbol{b}_1^{*\prime}\boldsymbol{R}_{22}\boldsymbol{b}_1^{*}=1$, the eigenvector of $\boldsymbol{R}_{22}^{-1}\boldsymbol{R}_{21}\boldsymbol{R}_{11}^{-1}\boldsymbol{R}_{12}$ corresponding to the eigenvalue ρ_1^2 is $\boldsymbol{b}_1^{*}=\left[2(1+\gamma)\right]^{-1/2}\mathbf{1}$.

Therefore, the first pair of canonical variables is:

$$\boldsymbol{a}_1^{*\prime}\boldsymbol{x}^{*}=\left[2(1+\alpha)\right]^{-1/2}\mathbf{1}'\boldsymbol{x}^{*},\ \boldsymbol{b}_1^{*\prime}\boldsymbol{y}^{*}=\left[2(1+\gamma)\right]^{-1/2}\mathbf{1}'\boldsymbol{y}^{*},$$

and the first canonical correlation is :

$$\rho_1=2|\beta|/\left[(1+\alpha)(1+\gamma)\right]^{1/2}.$$

Since $|\alpha|<1$, $|\gamma|<1$, so $\rho_1>|\beta|$, indicating that the first canonical correlation is larger than the correlation of two original sets of variables.

§ 10.3 Sample canonical correlations

- Suppose the data matrix is

$$(\boldsymbol{X} \quad \boldsymbol{Y}) = \begin{pmatrix} \boldsymbol{x}_1' & \boldsymbol{y}_1' \\ \vdots & \vdots \\ \boldsymbol{x}_n' & \boldsymbol{y}_n' \end{pmatrix} = \begin{pmatrix} x_{11} & \cdots & x_{1p} & y_{11} & \cdots & y_{1q} \\ \vdots & & \vdots & \vdots & & \vdots \\ x_{n1} & \cdots & x_{np} & y_{n1} & \cdots & y_{nq} \end{pmatrix}$$

then the sample covariance matrix is

$$\boldsymbol{S} = \begin{pmatrix} \boldsymbol{S}_{11} & \boldsymbol{S}_{12} \\ \boldsymbol{S}_{21} & \boldsymbol{S}_{22} \end{pmatrix}$$

$\boldsymbol{S}$ can be used as the estimation of $\boldsymbol{\Sigma}$.

When $n>p+q$, $\boldsymbol{S}_{11}^{-1}\boldsymbol{S}_{12}\boldsymbol{S}_{22}^{-1}\boldsymbol{S}_{21}$, $\boldsymbol{S}_{22}^{-1}\boldsymbol{S}_{21}\boldsymbol{S}_{11}^{-1}\boldsymbol{S}_{12}$ can be used as the estimation of $\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$, $\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}$; Their none-zero eigenvalue $r_1^2 \ge r_2^2 \ge \cdots \ge r_m^2$ can be used as the estimation of $\rho_1^2 \ge \rho_2^2 \ge \cdots \ge \rho_m^2$

- Corresponding eigenvector $\hat{\boldsymbol{a}}_1, \hat{\boldsymbol{a}}_2, \cdots, \hat{\boldsymbol{a}}_m$ are the estimation of $\boldsymbol{a}_1, \boldsymbol{a}_2, \cdots, \boldsymbol{a}_m$, $\hat{\boldsymbol{b}}_1, \hat{\boldsymbol{b}}_2, \cdots, \hat{\boldsymbol{b}}_m$ are the estimation of $\boldsymbol{b}_1, \boldsymbol{b}_2, \cdots, \boldsymbol{b}_m$. Motivated by **Proposition 1**, we can easily obtain $\hat{\boldsymbol{a}}_1, \hat{\boldsymbol{a}}_2, \cdots, \hat{\boldsymbol{a}}_m$ and $\hat{\boldsymbol{b}}_1, \hat{\boldsymbol{b}}_2, \cdots, \hat{\boldsymbol{b}}_m$.
- The positive square root r_j of r_i^2 is called the i-th sample canonical correlation. $\hat{\boldsymbol{a}}_i'\boldsymbol{x}$, $\hat{\boldsymbol{b}}_i'\boldsymbol{y}$ are called the i-th pair of sample canonical variates, $i = 1, 2, \cdots, m$.

Proposition 1 (Exer 1):

- $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \cdots, \boldsymbol{\alpha}_m$ are orthonormal eigenvectors of $\Sigma_{11}^{-1/2}\Sigma_{12}\Sigma_{22}^{-1/2}\Sigma_{21}$ corresponding to $\rho_1^2, \rho_2^2, \cdots, \rho_m^2$.
- $\boldsymbol{a}_1, \boldsymbol{a}_2, \cdots, \boldsymbol{a}_m$ are eigenvectors of $\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$ corresponding to $\rho_1^2, \rho_2^2, \cdots, \rho_m^2$.
- $\boldsymbol{b}_1, \boldsymbol{b}_2, \cdots, \boldsymbol{b}_m$ are eigenvectors of $\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}$ corresponding to $\rho_1^2, \rho_2^2, \cdots, \rho_m^2$.

The centered m pair canonical variates are:

$$u_i = \hat{a}_i'(x - \bar{x}), v_i = \hat{b}_i'(y - \bar{y}), i = 1,2,\cdots,m$$

Let the sample be $(\boldsymbol{x}_j, \boldsymbol{y}_j)$, j=1,2,⋯,n ,we have:

$$u_{ji} = \hat{\boldsymbol{a}}_i'\left(\boldsymbol{x}_j - \overline{\boldsymbol{x}}\right), \quad v_{ji} = \hat{\boldsymbol{b}}_i'\left(\boldsymbol{y}_j - \overline{\boldsymbol{y}}\right)$$

$$j = 1, 2, \cdots, n, \ \ i = 1, 2, \cdots, m$$

we call u_{ji} and v_{ij} the score of i -th sample canonical variates of (the j-th sample of) $\boldsymbol{x}_j$ and $\boldsymbol{y}_j$. Under constrain $\hat{\boldsymbol{a}}_i'\boldsymbol{S}_{11}\hat{\boldsymbol{a}}_i = 1$, the sample variance of u_i is

$$\frac{1}{n-1}\sum_{j=1}^{n} u_{ji}^2 = \frac{1}{n-1}\hat{\boldsymbol{a}}_i'\sum_{j=1}^{n}\left(\boldsymbol{x}_j - \overline{\boldsymbol{x}}\right)\left(\boldsymbol{x}_j - \overline{\boldsymbol{x}}\right)'\hat{\boldsymbol{a}}_i$$

$$= \hat{\boldsymbol{a}}_i'\boldsymbol{S}_{11}\hat{\boldsymbol{a}}_i = 1, \quad i = 1, 2, \cdots, m$$

- Similarly, the sample variance of v_i is given by

$$\frac{1}{n-1}\sum_{j=1}^{n} v_{ji}^2 = 1, \quad i = 1, 2, \cdots, m$$

- We can draw the scatterplot of scores of the first pair of canonical variates (u_{j1}, v_{j1}), $j=1,2,\cdots,n$, which maximizes the correlation between two sets of variables and can also be used to check for outliers. If needed, a second or more scatterplots of typical variables can be drawn.
- The pair of sample canonical variates (under the constraints above) maximizes the sample correlation, not the (population) correlation; the sample canonical variates within group are sample uncorrelated, not (population) uncorrelated ; the sample variance of the sample canonical variable is 1, while the (population) variance is not 1.

Example 10.3.1 A rehabilitation club (康复俱乐部) measured three physiological indicators (生理指标)- weight (x_1), waist circumference (x_2), and pulse (x_3) - and three training indicators - number of pull-ups (引体向上 y_1), number of sit-ups (起坐次数y_2), and number of jumps (y_3) - for 20 middle-aged individuals. The data is listed in Table 10.3.1. The data is given in table 10.3.1. Try to conduct canonical correlation analysis

Table 10. 3. 1 Physiological index and training index of a rehabilitation club

Number	x_1	x_2	x_3	y_1	y_2	y_3
1	191	36	50	5	162	60
2	189	37	52	2	110	60
3	193	38	58	12	101	101
4	162	35	62	12	105	37
5	189	35	46	13	155	58
6	182	36	56	4	101	42
7	211	38	56	8	101	38

8	167	34	60	6	125	40
9	176	31	74	15	200	40
10	154	33	56	17	251	250
11	169	34	50	17	120	38
12	166	33	52	13	210	115
13	154	34	64	14	215	105
14	247	46	50	1	50	50
15	193	36	46	6	70	31
16	202	37	62	12	210	120
17	176	37	54	4	60	25
18	157	32	52	11	230	80
19	156	33	54	15	225	73
20	138	33	68	2	110	43

Solution: By calculation, we obtain

$$\hat{\boldsymbol{R}}_{11} = \begin{pmatrix} 1 & & \\ 0.870 & 1 & \\ -0.366 & -0.353 & 1 \end{pmatrix}, \quad \hat{\boldsymbol{R}}_{22} = \begin{pmatrix} 1 & & \\ 0.696 & 1 & \\ 0.496 & 0.669 & 1 \end{pmatrix}$$

$$\hat{\boldsymbol{R}}_{12} = \hat{\boldsymbol{R}}'_{21} = \begin{pmatrix} -0.390 & -0.493 & -0.226 \\ -0.552 & -0.646 & -0.192 \\ 0.151 & 0.225 & 0.035 \end{pmatrix}$$

The eigenvalue of $\hat{\boldsymbol{R}}_{11}^{-1}\hat{\boldsymbol{R}}_{12}\hat{\boldsymbol{R}}_{22}^{-1}\hat{\boldsymbol{R}}_{21}$ is 0.6330, 0.0402 and 0.0053, then

$$r_1=0.796, \quad r_2=0.201, \quad r_3=0.073$$

The sample canonical correlation coefficient vectors of $\hat{\boldsymbol{R}}_{11}^{-1}\hat{\boldsymbol{R}}_{12}\hat{\boldsymbol{R}}_{22}^{-1}\hat{\boldsymbol{R}}_{21}$ and $\hat{\boldsymbol{R}}_{22}^{-1}\hat{\boldsymbol{R}}_{21}\hat{\boldsymbol{R}}_{11}^{-1}\hat{\boldsymbol{R}}_{12}$ corresponding to r_1, r_2, r_3 are given respectively by:

$$\hat{\boldsymbol{a}}_1^* = \begin{pmatrix} -0.775 \\ 1.579 \\ -0.059 \end{pmatrix}, \quad \hat{\boldsymbol{a}}_2^* = \begin{pmatrix} -1.884 \\ 1.181 \\ -0.231 \end{pmatrix}, \quad \hat{\boldsymbol{a}}_3^* = \begin{pmatrix} -0.191 \\ 0.506 \\ 1.051 \end{pmatrix}$$

$$\hat{\boldsymbol{b}}_1^* = \begin{pmatrix} -0.350 \\ -1.054 \\ 0.716 \end{pmatrix}, \quad \hat{\boldsymbol{b}}_2^* = \begin{pmatrix} -0.376 \\ 0.124 \\ 1.062 \end{pmatrix}, \quad \hat{\boldsymbol{b}}_3^* = \begin{pmatrix} -1.297 \\ 1.237 \\ -0.419 \end{pmatrix}$$

Therefore, the first pair of sample canonical variates is:

$$u_1^* = -0.775x_1^* + 1.579x_2^* - 0.059x_3^*$$

$$v_1^* = -0.350y_1^* - 1.054y_2^* + 0.716y_3^*$$

If needed ,the second pair of sample canonical variates is:

$$u_2^* = -1.884x_1^* + 1.181x_2^* - 0.231x_3^*$$

$$v_2^* = -0.376y_1^* + 0.124y_2^* + 1.062y_3^*$$

Example 10.3.2 As part of investigating the impact of organizational structure on "job satisfaction," Dunham conducted a survey to assess the degree of correlation between job satisfaction and occupational characteristics. Measurements were taken from n=784 administrative personnel who transitioned from various branches of a large retail company. Five occupational characteristic variables (use feedback (x_1), task importance (x_2), task diversity (x_3), task characteristics (x_4), autonomy (x_5)), and seven job satisfaction metrics (supervisor satisfaction (y_1), career prospects satisfaction (y_2), financial satisfaction (y_3), workload satisfaction (y_4), company status satisfaction (y_5), job type satisfaction (y_6), and overall satisfaction (y_7))were assessed. The sample correlation matrix for the 784 subjects is as follows:

$$\hat{\boldsymbol{R}}_{11} = \begin{pmatrix} 1.00 & & & & \\ 0.49 & 1.00 & & & \\ 0.53 & 0.57 & 1.00 & & \\ 0.49 & 0.46 & 0.48 & 1.00 & \\ 0.51 & 0.53 & 0.57 & 0.57 & 1.00 \end{pmatrix}$$

$$\hat{\boldsymbol{R}}_{22} = \begin{pmatrix} 1.00 & & & & & & \\ 0.43 & 1.00 & & & & & \\ 0.27 & 0.33 & 1.00 & & & & \\ 0.24 & 0.26 & 0.25 & 1.00 & & & \\ 0.34 & 0.54 & 0.46 & 0.28 & 1.00 & & \\ 0.37 & 0.32 & 0.29 & 0.30 & 0.35 & 1.00 & \\ 0.40 & 0.58 & 0.45 & 0.27 & 0.59 & 0.31 & 1.00 \end{pmatrix}$$

$$\hat{\boldsymbol{R}}_{12} = \hat{\boldsymbol{R}}'_{21} = \begin{pmatrix} 0.33 & 0.32 & 0.20 & 0.19 & 0.30 & 0.37 & 0.21 \\ 0.30 & 0.21 & 0.16 & 0.08 & 0.27 & 0.35 & 0.20 \\ 0.31 & 0.23 & 0.14 & 0.07 & 0.24 & 0.37 & 0.18 \\ 0.24 & 0.22 & 0.12 & 0.19 & 0.21 & 0.29 & 0.16 \\ 0.38 & 0.32 & 0.17 & 0.23 & 0.32 & 0.36 & 0.27 \end{pmatrix}$$

The canonical correlation coefficient vectors are given in Table 10.3.2

Table 10.3.2 Canonical variate coefficient and canonical correlations

Standardized variables	$\hat{\boldsymbol{a}}_1^*$	$\hat{\boldsymbol{a}}_2^*$	$\hat{\boldsymbol{a}}_3^*$	$\hat{\boldsymbol{a}}_4^*$	$\hat{\boldsymbol{a}}_5^*$
x_1^*	0.42	0.34	−0.86	−0.79	0.03
x_2^*	0.20	−0.67	0.44	−0.27	0.98
x_3^*	0.17	−0.85	−0.26	0.47	−0.91
x_4^*	−0.02	0.36	−0.42	1.04	0.52
x_5^*	0.46	0.73	0.98	−0.17	−0.44
r_j	0.55	0.24	0.12	0.07	0.06
Standardized variables	$\hat{\boldsymbol{b}}_1^*$	$\hat{\boldsymbol{b}}_2^*$	$\hat{\boldsymbol{b}}_3^*$	$\hat{\boldsymbol{b}}_4^*$	$\hat{\boldsymbol{b}}_5^*$
y_1^*	0.43	−0.09	0.49	−0.13	−0.48
y_2^*	0.21	0.44	−0.78	−0.34	−0.75
y_3^*	−0.04	−0.09	−0.48	−0.61	0.35
y_4^*	0.02	0.93	−0.01	0.40	0.31
y_5^*	0.29	−0.10	0.28	−0.45	0.70
y_6^*	0.52	−0.55	−0.41	0.69	0.18
y_7^*	−0.11	−0.03	0.93	0.27	−0.01

The first pair of canonical variables is

$$u_1^* = 0.42x_1^* + 0.20x_2^* + 0.17x_3^* - 0.02x_4^* + 0.46x_5^*$$

$$v_1^* = 0.43y_1^* + 0.21y_2^* - 0.04y_3^* + 0.02y_4^* + 0.29y_5^* + 0.52y_6^* - 0.11y_7^*$$

- According to the canonical coefficients, u_1^* primarily represents the variables of user feedback and autonomy (自主权), with the three task variables appearing to be less important.
- v_1^* primarily represents the variables of supervisor satisfaction and job type satisfaction, followed by career prospects satisfaction and company status satisfaction variables.
- We can also analyze the canonical variables from the perspective of correlation coefficients. The sample correlation coefficients between the original variables and the first pair of canonical variables are listed in Table 10.3.3.

Table 10.3.3 Sample correlations between original variables and canonical variables

Original variables x	Sample canonical variates u_1^*	v_1^*	Original variables y	Sample canonical variates u_1^*	v_1^*
x_1 : use feedback	0.83	0.46	y_1: supervisor satisfaction	0.42	0.76
x_2: task importance	0.73	0.40	y_2 : career prospects satisfaction	0.36	0.64
x_3: task diversity	0.75	0.42	y_3: financial satisfaction	0.21	0.39
x_4: task characteristics	0.62	0.34	y_4: workload satisfaction	0.21	0.38
x_5: autonomy	0.86	0.48	y_5 : company status satisfaction	0.36	0.65
			y_6 : job type satisfaction	0.45	0.80
			y_7 : overall satisfaction	0.28	0.50

- Five jobs characteristic variables have roughly the same correlations with the first canonical variate u_1^*. We may interpret u_1^* as a job characteristic index.
- v_1^* primary represents supervisor satisfaction, career prospects satisfaction, company status satisfaction, and job type satisfaction.
- v_1^* can be interpreted as job satisfaction - company status variable, which is essentially consistent with the interpretation based on canonical coefficients.
- The sample correlation coefficient between the first pair of canonical variables u_1^* and v_1^* denoted as r_1, is 0.55, indicating a certain level of correlation between occupational characteristics and job satisfaction.

§ 10.4 Testing the significance of the canonical correlations

- 1. Testing whether all the population canonical correlation coefficients are zero
- 2. Testing whether part of the population canonical correlation coefficients are zero

1. Testing whether all the population canonical correlation coefficients are zero

- Suppose $\begin{pmatrix} \boldsymbol{x} \\ \boldsymbol{y} \end{pmatrix} \sim N_{p+q}(\mu, \boldsymbol{\Sigma}), \boldsymbol{\Sigma} > 0$. Suppose $S = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix}$ is sample covariance matrix, and $n>p+q$.
- Consider the hypotheses ：

$$H_0: \ \rho_1=\rho_2=\cdots=\rho_m=0$$
$$H_1: \ \rho_1, \rho_2, \cdots, \rho_m \text{ are not all } 0 \qquad (10.4.1)$$

where m=min$\{p, q\}$. If the test accept $H_{0,}$ then it is meaningless to discuss the correlation of two sets of variables. If the test reject H_0 , then the first pair of canonical variables is assumed to be significant.

- (10.4.1) is equivalent to the hypotheses

$$H_0: \ \boldsymbol{\Sigma}_{12}=\mathbf{0}, \quad H_1: \ \boldsymbol{\Sigma}_{12}\neq\mathbf{0}$$

H_0 holds means that $\boldsymbol{x}$ and $\boldsymbol{y}$ are uncorrelated.

- Likelihood ratio statistics is

$$\Lambda_1 = \prod_{i=1}^{m}\left(1 - r_i^2\right)$$

where $r_1^2, r_2^2, \cdots, r_m^2$ are the nonzero eigenvalues of $S_{11}^{-1}S_{12}S_{22}^{-1}S_{21}$.

- For sufficient large n, when H_0 holds, the statistics

$$Q_1 = -\left[n - \frac{1}{2}(p+q+3)\right]\ln\Lambda_1 \dot{\sim} \chi^2(pq)$$

- Under given α, if $Q_1 \geq \chi_\alpha^2(pq)$, then we reject H_0, assuming that the correlation between canonical variates are significant; otherwise, we assume that the first canonical correlation coefficient is not significant.

Example 10.4.1 In example 10.3.1, suppose we have multivariate normal data, and want to test

$$H_0:\ \rho_1=\rho_2=\rho_3=0,\ H_1:\ \rho_1\neq 0,$$

its likelihood ratio statistics is given by

$$\Lambda_1=\left(1-r_1^2\right)\left(1-r_2^2\right)\left(1-r_3^2\right)$$

The eigenvalue of $\hat{R}_{11}^{-1}\hat{R}_{12}^{-1}\hat{R}_{22}^{-1}\hat{R}_{21}$ is 0.6330, 0.0402 and 0.0053.

$$=\left(1-0.6330\right)\left(1-0.0402\right)\left(1-0.0053\right)=0.3504$$

$$Q_1=-\left[20-\frac{1}{2}\left(3+3+3\right)\right]\ln\Lambda_1=-15.5\times\ln 0.3504=16.255$$

χ^2 distribution table gives that $\chi^2_{0.10}(9)=14.684$, $\chi^2_{0.05}(9)=16.919$

Therefore, under the significance level α=0.10, we reject H_0, that is, we assume that at least one of the canonical correlation is significant (p=0.062).

2. Testing whether part of the population canonical correlation coefficients are zero

- If $H_0:\ \rho_1=\rho_2=\cdots=\rho_m=0$ is rejected, then we should further test the hypothesis

$$H_0:\ \rho_2=\cdots=\rho_m=0$$

$$H_1:\ \rho_2,\cdots,\rho_m \text{ are not all } 0$$

If the null hypothesis H_0 is accepted, then we think that only the first pair of canonical variables is significant; if the null hypothesis H_0 is rejected, then we assume that the second pair of canonical variables is significant.

- So forth until some k, suppose $H_0:\ \rho_{k+1}=\cdots=\rho_m=0$ is accepted, then we can assume that only first k pair of canonical variables are significant.
- For the hypothesis:

$$H_0:\ \rho_{k+1}=\cdots=\rho_m=0$$

$$H_1:\ \rho_{k+1},\cdots,\rho_m \text{ are not all } 0$$

The statistics is given by:

$$\Lambda_{k+1} = \prod_{i=k+1}^{m} \left(1 - r_i^2\right)$$

For sufficiently large *n*, when H_0 is true, the statistics

$$Q_{k+1} = -\left[n - k - \frac{1}{2}(p+q+3) + \sum_{i=1}^{k} r_i^{-2}\right] \ln \Lambda_{k+1}$$

has the distribution χ^2 $[(p-k)(q-k)]$.

Given α, if $Q_{k+1} \geq \chi_\alpha^2\left[(p-k)(q-k)\right]$, then we reject H_0, assume that ρ_{k+1} is significant, that is, the (k+1)-canonical variables is correlated significantly.

- The testing above is actually a sequence of hypotheses, we test until some k where H_0 is not rejected. In fact, the overall significance level is not α, and would be difficult to determine.

Example 10.4.2 In example10.3.1, furthermore, if we want to test :

$$H_0:\ \rho_2=\rho_3=0,\quad H_1:\ \rho_2\neq 0$$

the statistics is given by

The eigenvalue of $\hat{R}_{11}^{-1}\hat{R}_{12}^{-1}\hat{R}_{22}^{-1}\hat{R}_{21}$ is 0.6330, 0.0402 and 0.0053.

$$\Lambda_2=\left(1-r_2^2\right)\left(1-r_3^2\right)=\left(1-0.0402\right)\left(1-0.0053\right)=0.9547$$

$$Q_2=-\left[20-1-\frac{1}{2}\left(3+3+3\right)+r_1^{-2}\right]\ln\Lambda_2$$

$$=-16.08\times\ln 0.9547=0.745<7.779=\chi_{0.10}^2\left(4\right)$$

So we accept H_0, that is, we think that the second canonical correlations is not significant (p=0.946). Therefore, only one canonical correlations are significant.

10. 5 Application of canonical correlation analysis

"短视频观看行为与心理健康的关系研究"

1 研究背景与科学问题

在数字时代，短视频平台（如TikTok、抖音）的流行引发了人们对心理健康影响的广泛关注。传统研究通常仅关注使用时长与单一心理健康指标（如抑郁评分）的关系，但忽略了：

1. 观看行为的多样性（内容类型，观看时长，观看时段，互动方式等）
2. 心理健康的多维性（情绪、焦虑、幸福感等）

典型相关性分析（CCA） 能够揭示两组高维变量（观看行为 vs. 心理健康）之间的潜在关联模式，比简单的回归分析提供更全面的见解。

2. 数据准备与预处理

(1) 数据收集与变量定义

我们采集了500名18-25岁短视频用户的数据，包括：

- **短视频行为数据**（通过API获取30天记录）
- **心理健康量表数据**（问卷调查）

10. 5 Application of canonical correlation analysis

"短视频观看行为与心理健康的关系研究"

短视频观看行为变量（X组）：

变量名	说明	处理方式
watch_time (X_1)	日均观看分钟数	连续变量
Entertainment video viewing ratio (X_2)	娱乐视频观看比例	百分比转换
like_ratio (X_3)	点赞视频占比	百分比转换
night_use (X_4)	23:00-5:00使用占比	百分比转换
edu_content (X_5)	教育类内容占比	百分比转换

心里健康变量（Y组）：

变量名	量表	维度
PHQ9 (Y_1)	抑郁症状量表	0-27分
GAD7 (Y_2)	广泛性焦虑量表	0-21分
SWLS (Y_3)	生活满意度量表	5-35分
UCLA (Y_4)	孤独感量表	20-80分
PSQI (Y_5)	睡眠质量	20-80分

10. 5 Application of canonical correlation analysis

"短视频观看行为与心理健康的关系研究"

(2) 数据预处理

1. 数据清洗：
   1. 剔除极端值（如每日观看时长 > 12 小时的用户）。
   2. 检查缺失值，采用插补或删除处理。
2. 标准化：
   1. 对 **X** 和 **Y** 变量分别进行 **Z-score 标准化**（均值为 0，标准差为 1），以消除量纲影响。
3. 相关性检查：
   1. 计算 **X 组内** 和 **Y 组内** 的 Pearson 相关性，确保变量间存在一定关联（避免完全独立变量影响 CCA 效果）。

10. 5 Application of canonical correlation analysis

"短视频观看行为与心理健康的关系研究"

(2) 数据预处理：

4. 采用置换检验(permutation test)评估典型相关分析的显著性(python)：

```
from sklearn.utils import shuffle
null_dist = []
for _ in range(1000):
Y_shuffled = shuffle(Y_scaled)
cca_null = CCA(n_components=1).fit(X_scaled, Y_shuffled)
U_null, V_null = cca_null.transform(X_scaled, Y_shuffled)
null_dist.append(np.corrcoef(U_null.T, V_null.T)[0,1])
p_value = (np.sum(null_dist >= corr1) + 1) / (1000 + 1)
```

10. 5 Application of canonical correlation analysis

"短视频观看行为与心理健康的关系研究"

- **3. 典型相关性分析（CCA）步骤**

1. **计算协方差矩阵**：
   1. 计算 **X** 和 **Y** 的样本相关矩阵R_{11}, R_{22}，以及交叉样本相关矩阵 R_{12}。
2. **求解典型变量**：
   1. 计算矩阵 $S_{11}^{-1}S_{12}S_{22}^{-1}S_{21}$和$S_{22}^{-1}S_{21}S_{11}^{-1}S_{12}$ 的特征值$\rho_1^2, \rho_2^2, \cdots, \rho_m^2$和对应的特征向量$\boldsymbol{a}_i, \boldsymbol{b}_i$。
   2. 选取前 **k** 个显著的典型相关系数 $\rho_1, \rho_2, \cdots, \rho_k$ （通常通过 Bartlett 检验或交叉验证确定）。
3. **计算典型载荷（Canonical Loadings）**：

   计算 **X** 和 **Y** 变量在典型变量上的载荷，解释各变量对典型相关的贡献。
4. 显著性检验

   使用Bartlett近似卡方检验检查典型相关系数是否显著($p < 0.05$)。

10. 5 Application of canonical correlation analysis

"短视频观看行为与心理健康的关系研究"

(1) 第一典型变量对（ρ=0.47, p<0.001）

观看行为	载荷	心理健康	载荷
watch_time	0.82	PHQ9(抑郁症状量表)	0.78
night_use	0.65	GAD7(广泛性焦虑量表)	0.71
Like_ratio	-0.02	PSQL (睡眠质量)	0.46
Entertainment video viewing ratio	0.65	UCLA(孤独感量表)	0.63
edu_content	-0.32	SWLS(生活满意度量表)	-0.59

行为模式解释：

高载荷变量组合揭示：

- 夜间过度观看娱乐类短视频的用户，抑郁和焦虑水平较高。
- 与教育内容观看呈负相关

心理关联：

强烈关联抑郁/焦虑症状和孤独感，与生活满意度负相关

10. 5 Application of canonical correlation analysis

"短视频观看行为与心理健康的关系研究"

(2) 第二典型变量对（ρ=0.32, p=0.012）

观看行为	载荷	心理健康	载荷
like_ratio	0.68	SWLS(生活满意度量表)	0.54
edu_content	0.61	UCLA (孤独感量表)	-0.47

行为模式解释：

- **主动选择性使用：**偏好知识类内容并积极互动的用户，幸福感和社交满意度较高。
- 与生活满意度正相关，孤独感负相关

10. 5 Application of canonical correlation analysis

"短视频观看行为与心理健康的关系研究"

4. 创新性发现与启示

•夜间娱乐短视频观看可能对心理健康不利，建议平台优化“防沉迷”机制。

•知识类内容和积极互动可能提升幸福感，可鼓励高质量内容创作。

•CCA 比单变量分析更全面，能发现行为模式与心理状态的复杂关联。

10. 5 Application of canonical correlation analysis

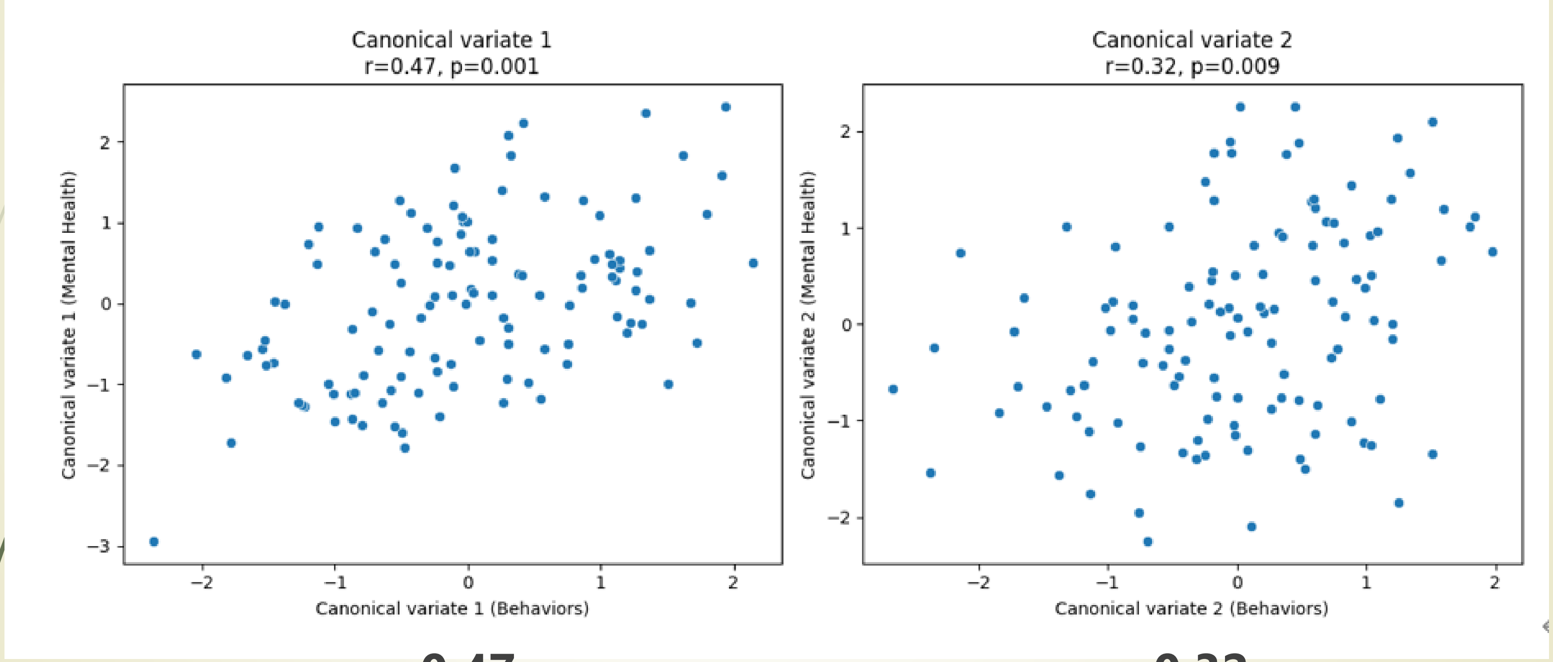
"短视频观看行为与心理健康的关系研究"

代码(python)示例

```
from sklearn.cross_decomposition import CCA
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
# 假设 X (观看行为) 和 Y (心理健康) 是标准化后的数据
X = np.random.randn(500, 5) # 模拟数据
Y = np.random.randn(500, 5)
# 拟合 CCA
cca = CCA(n_components=2)
cca.fit(X, Y)
# 计算典型变量
 U, V = cca.transform(X, Y)
 # 计算典型相关系数
corr = [np.corrcoef(U[:, i], V[:, i])[0, 1] for i in range(2)]
print("典型相关系数:", corr)
# 绘制典型变量散点图
 plt.scatter(U[:, 0], V[:, 0], alpha=0.5)
 plt.xlabel("短视频行为典型变量 U1")
plt.ylabel("心理健康典型变量 V1")
plt.title("第一对典型变量关系")
plt.show()
```

10. 5 Application of canonical correlation analysis

"短视频观看行为与心理健康的关系研究"



ρ=0.47 **ρ=0.32**

Figure 1 Visualization: Canonical Variates Scatterplots